

Revised Load Interconnection System Impact Study Report

For Queue Project Q18-1112

New Hollandale 230 kV
October 18, 2019

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q18-1112. The Interconnection Customer proposes a 246.9 MW large load (data center) facility interconnecting at the New Hollandale 230 kV substation in the Ohio area. The request was received on November 24, 2018. The study evaluated system intact and single contingency conditions using the regional planning case, including 465 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 3 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$2,859,000.

3. Network Impacts

3.1 Load Deliverability Analysis

1. Facility: Staunton / Piqua Power Plant 230 KV
Contingency: East Methuen 500 kV/230 kV Autotransformer
Loading: 129.8% (Rating: 517 MVA)
DFAX: 6.9%
Pre-project loading: 126.5%
2. Facility: Carley Road / Piqua Power Plant 230 KV L
Contingency: E Methuen 500/230kV XFMR
Loading: 124.4% (Rating: 540 MVA)
DFAX: 6.9%
Pre-project loading: 121.2%
3. Facility: Baraga Vlg / Saunton 230 KV
Contingency: East Methuen 500/230 kV Transformer
Loading: 121.7% (Rating: 532 MVA)
DFAX: 6.9%
Pre-project loading: 118.5%

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Staunton / Piqua Power Plant 230 KV	Rebuild line with higher-capacity conductor	\$22,001,000	3% (\$629,000)
Carley Road / Piqua Power Plant 230 KV L	Rebuild line with higher-capacity conductor	\$45,741,000	3% (\$1,308,000)
Baraga Vlg / Saunton 230 KV	Rebuild line with higher-capacity conductor	\$32,240,000	3% (\$922,000)
	Total allocated to project		\$2,859,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.